

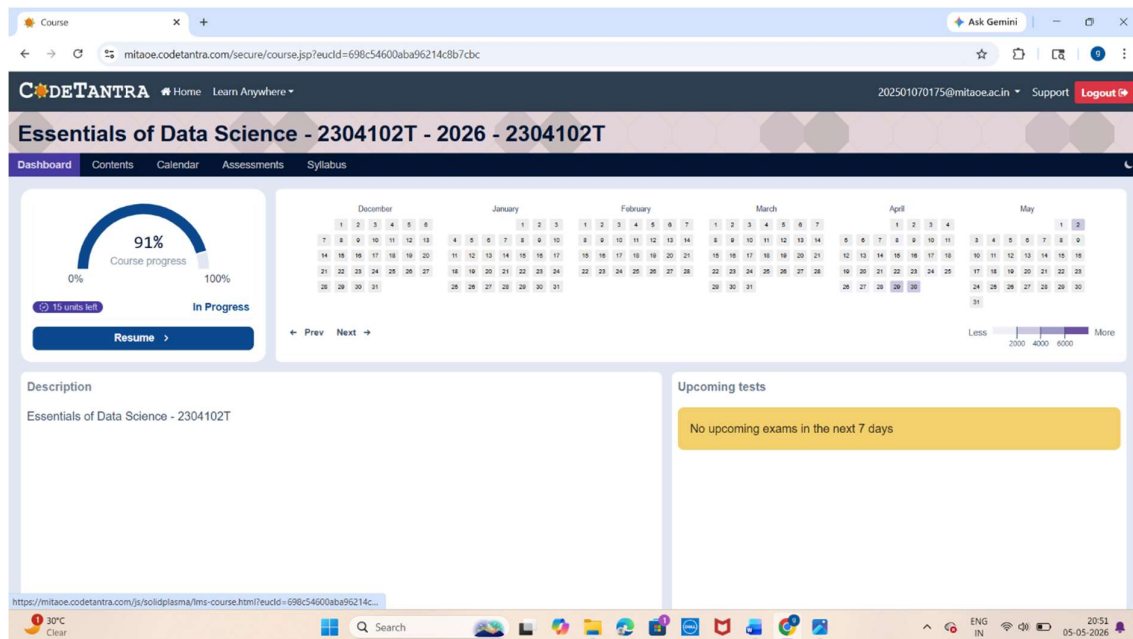
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This certificate is awarded to

Gauri Gawade

for successfully completing

Python Essentials 1

offered by MIT Academy of Engineering, alandi Pune
through the Cisco Networking Academy program.

Anuradha Pawar

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1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

$1 \leq n \leq 999$

Input Format:

The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2 n=int(input())
3 if 1<=n<=9:
4     print(n**2)
5 elif 10<=n<=99:
6     print(f"{n**0.5:.2f}")
7 elif 100<=n<=999:
8     print(f"{n**(1/3):.2f}")
9 else:
10    print("Invalid")
```

TerminalTest cases

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:

An integer representing the number of days between the given dates.

Note:

The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import datetime
2
3 date1_str = input()
4 date2_str = input()
5
6 d1 = datetime.strptime(date1_str, "%Y-%m-%d")
7 d2 = datetime.strptime(date2_str, "%Y-%m-%d")
8
9 difference = d2 - d1
10
11 print(difference.days)
```

TerminalTest cases

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n=int(input())
2 r=0
3 while n!=0:
4     d=n%10
5     r=r*10+d
6     n=n//10
7 print(r)
```

1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 n=int(input())
2 for i in range(1,11):
3     print(f"{n} x {i} = {n*i}")
4
5
6
```

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:
If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

passorFa...

1n = int(input())
2marks = list(map(int, input().split()))
3if any(m < 40 for m in marks):
4 print("Fail")
5else:
6 total_marks = sum(marks)
7 aggregate = total_marks / n
8 if aggregate > 75:
9 grade = "Grade: Distinction"
10 elif aggregate >= 60:
11 grade = "Grade: First Division"
12 elif aggregate >= 50:
13 grade = "Grade: Second Division"
14 elif aggregate >= 40:
15 grade = "Grade: Third Division"
16 print(f"Aggregate Percentage: {aggregate:.2f}")
17 print(grade)

TerminalTest cases

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.
Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

1def fibonacci(n):
2 if n == 1:
3 return 0
4 elif n == 2:
5 return 1
6 else:
7 return fibonacci(n-1) + fibonacci(n-2)
8 n = int(input())
9 for i in range(1, n + 1):
10 print(fibonacci(i), end=" ")
11

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

1 n = int(input())
2 for i in range(1,n + 1):
3 print("*" * i)

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

1 n=int(input())
2 for i in range(1,n+1):
3 for j in range(1,i+1):
4 print(j,end=" ")
5 print()

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2.1.1. List operations

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:
1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:
• **Add:** Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
• **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
• **Display:** Displays the current list of integers. If the list is empty, display "List is empty".
• **Quit:** Exits the program.
• The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).

Sample Test Cases

listOps.py

```
1 list=[]
2 while True:
3     print("1. Add")
4     print("2. Remove")
5     print("3. Display")
6     print("4. Quit")
7     a=int(input("Enter choice: "))
8     if a==1:
9         v=input("Integer: ")
10        if(v.isdigit()==True):
11            list.append(int(v))
12            print("List after adding:",list)
13        else:
14            print("Invalid input")
15    elif a==2:
16        if (len(list)==0):
17            print("List is empty")
18        else:
19            f=int(input("Integer: "))
20            if f in list:
21                list.remove(int(f))
22                print("List after removing:",list)
23            else:
24                print("Element not found")
25    elif a==3:
```

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2.1.2. Dictionary Operations

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:
1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:
1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.
The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:
• All operations must be performed using dictionary methods.
• Perform deletion only if possible; leave the dictionary unchanged.
• Refer to the visible test cases for better understanding and strictly match with the input/outputs.

DictOper...

```
1 # Initial dictionary with 10 predefined records
2 student = {
3     1: "Amit",
4     2: "Riya",
5     3: "Kiran",
6     4: "Neha",
7     5: "Arjun",
8     6: "Pooja",
9     7: "Rahul",
10    8: "Sneha",
11    9: "Vikram",
12    10: "Anjali"
13 }
14 print("Original Dictionary:",student)
15
16
17 key=int(input())
18 value=input()
19 student[key]=value
20 print("After Insertion:",student)
21
22
23 key=int(input())
24 value=input()
25 student[key]=value
```

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2.2.1. Linear search Technique

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print **Not found**.

Sample Test Case:
Input:
1 2 3 4 3 5 6
3
Output:
2

Sample Test Cases

CTP1709...

1 arr = list(map(int,input().split()))
2 key = int(input())
3 if key in arr:
4 print(arr.index(key))
5 else:
6 print("Not found")

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Test cases

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2.2.2. Captain of the Team

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format
The output should be the height (in centimeters) of the tallest player.

Sample Test Cases

captainof...

1 heights = list(map(int,input().split()))
2 print(max(heights))

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Test cases

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`.
- The shape of the array using `shape`.
- The total number of elements in the array using `size`.

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

1 `import numpy as np`

2

3 `r,c =map(int,input().split())`

4

5 `data = []`

6 `for i in range(r):`

7 `data.extend(map(int,input().split()))`

8

9 `arr = np.array(data).reshape(r,c)`

10

11 `print(arr)`

12 `print(arr.ndim)`

13 `print(arr.shape)`

14 `print(arr.size)`

15

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3.2.1. Numpy: Matrix Operations

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

- Addition (`matrix_a + matrix_b`)
- Subtraction (`matrix_a - matrix_b`)
- Element-wise Multiplication (`matrix_a * matrix_b`)
- Matrix Multiplication (`matrix_a . matrix_b`)
- Transpose of Matrix A

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

- The result of Addition.
- The result of Subtraction.
- The result of Element-wise Multiplication.
- The result of Matrix Multiplication.
- The Transpose of Matrix A.

matrixOp...

1 `import numpy as np`

2

3 `# Input matrices`

4 `print("Enter Matrix A:")`

5 `matrix_a = np.array([list(map(int, input().split())) for i in range(3)])`

6

7 `print("Enter Matrix B:")`

8 `matrix_b = np.array([list(map(int, input().split())) for i in range(3)])`

9

10 `#Addition`

11 `madd=matrix_a+matrix_b`

12 `print("Addition (A + B):")`

13 `print(madd)`

14 `#Subtraction`

15 `msub=matrix_a-matrix_b`

16 `print("Subtraction (A - B):")`

17 `print(msub)`

18 `# Multiplication (element-wise)`

19 `mmul=matrix_a*matrix_b`

20 `print("Element-wise Multiplication (A * B):")`

21 `print(mmul)`

22 `# Matrix multiplication (dot product)`

23 `mdot=np.dot(matrix_a,matrix_b)`

24 `print("A dot B:")`

25 `print(mdot)`

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 print("Horizontal Stack:")
12 harr = np.hstack((arr1, arr2))
13 print(harr)
14
15 # Perform vertical stacking (vstack)
16 print("Vertical Stack:")
17 varr = np.vstack((arr1, arr2))
18 print(varr)
19
20
21
```

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3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

```
1 import numpy as np
2
3 # Take user input for the start, stop, and step of the sequence
4 start = int(input())
5 stop = int(input())
6 step = int(input())
7
8 # Generate the sequence using np.arange()
9 a = np.arange(start, stop, step)
10 print(a)
11 # Print the generated sequence
12
13
```

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operations

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

- Arithmetic Operations:**
 - Compute the element-wise sum, difference, and product of the two arrays.
- Statistical Operations:**
 - Calculate the mean, median, and standard deviation of array A.
- Bitwise Operations:**
 - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4     # Convert A and B to NumPy arrays
5     A=np.array(A)
6     B=np.array(B)
7     # Arithmetic Operations
8     sum_result = A+B
9     diff_result = A-B
10    prod_result = A*B
11
12    # Statistical Operations
13    mean_A = np.mean(A)
14    median_A = np.median(A)
15    std_dev_A = np.std(A)
16
17    # Bitwise Operations
18    and_result = A&B
19    or_result = A|B
20    xor_result = A^B
21
22
23
24    # Output results with one space between each element
25    print("Element-wise Sum: ", ' '.join(map(str, sum_result)))
```

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3.2.5. Numpy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>
View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>
Copy array: <copy_array>

Sample Test Cases

copyAnd...

```
1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
14
15 # Modify the view
16 view_array[0] = 99
17 print("Original array after modifying view:", original_array)
18 print("View array:", view_array)
19
20 # Modify the copy
21 copy_array[1] = 88
22 print("Original array after modifying copy:", original_array)
23 print("Copy array:", copy_array)
24
```

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching:** Find the indices where search_value appears in array1 and print these indices.
- Counting:** Count how many times count_value appears in array1 and print the count.
- Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing array1.
- An integer search_value represents the value to search for in the array.
- An integer count_value represents the value to count in the array.
- An integer broadcast_value represents the value to add to each element of the array.

Output Format:

- The indices where search_value occurs in array1.
- The count of occurrences of count_value in array1.
- The array after adding the broadcast_value to each element.

arrayOpe...

```
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split())))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
9 broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
12 a=np.where(array1==search_value)[0]
13 print(a)
14 # Count occurrences in array1
15 b=np.count_nonzero(array1==count_value)
16 print(b)
17 # Broadcasting addition
18 c=array1+ broadcast_value
19 print(c)
20 # Sort the first array
21 d = np.sort(array1)
22 print(d)
```

Thunderstorm w...
In effect

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3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- Find total students:** Determine the total number of students in the dataset.
- Print all student roll numbers:** Extract and print the roll numbers of all students.
- Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- Print all subject marks:** Display the marks of all students for each subject.
- Find total marks of students:** Compute the total marks for each student across all subjects.
- Find the average marks of each student:** Compute the average marks for each student.
- Find average marks of each subject:** Compute the average marks for all students in each subject.
- Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- Find the count of students who not less than 40 marks in Subject 4:** Count the number of

Operatio...

```
1 import numpy as np
2
3 a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
4
5 # 1. Print all student details
6 print("All student Details:\n",a)
7
8 # 2. print total students
9
10 print("Total Students:", len(a))
11
12 # 3. Print all student Roll numbers
13 print("All Student Roll Nos", a[:,0]...)
14
15 # 4. Print subject 1 marks
16 print("Subject 1 Marks", a[:,1])
17
18 # 5. print minimum marks of Subject 2
19 print("Min marks in Subject 2", a[:,2].min()....)
20
21 # 6. print maximum marks of Subject 3
22 print("Max marks in Subject 3", a[:,3].max()....)
23
24 # 7. Print All subject marks
25 a1 = np.delete(a,0,1)
```

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4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep=";", header=None, names=["Name", "Age", "Grade"])
6
7
8 # write your code here..
9 print("First five rows:")
10 print(data.head())
11
12 # 2. Calculate average age (2 decimal places)
13 avg_age = round(data["Age"].mean(), 2)
14 print("Average age:", avg_age)
15
16 # 3. Filter students with grade up to 'B'
17 filtered = data[data["Grade"].isin(["A", "B"])]
18
19 print("Students with a grade up to B")
20 print(filtered)
21
```

Terminal

Test cases

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4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to lowercase

datafram...

```
1 # import pandas as pd
2
3 ## Provided dictionary of lists
4 # data = {
5 #     'Name': ['Alice', 'Bob', 'Charlie'],
6 #     'Age': [25, 30, 35],
7 # }
8
9 ## Convert the dictionary to a DataFrame
10 # df = pd.DataFrame(data)
11
12 ## Display the original DataFrame
13 # print("Original DataFrame:")
14 # print(df)
15
16 ## Adding a new row
17
18
19 ## Display the DataFrame after adding a new row
20 # print("After adding a row:\n",df)
21
22 ## Modifying a row
23
24
25 ## Display the DataFrame after modifying a row
```

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Test cases

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4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 s = pd.Series(numbers)
8 # Grouping by even and odd numbers and calculating the mean
9 grouped = s.groupby(s % 2 == 0).mean()
10
11 # Display the mean of even and odd numbers with labels
12 grouped.index = ['Even' if is_even else 'Odd' for is_even in
13                 grouped.index]
14 print("Mean of even and odd numbers:")
15 print(grouped)
```

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4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 # import pandas as pd
2
3 ## Prompt the user for the file name
4 file_name = input()
5
6 ## Load the data
7 df = pd.read_csv(file_name)
8
9 # d
10 ## Find the month with the highest total sales
11 #best_month =
12 #highest_sales =
13
14 # print(f"Best month: {best_month}")
15 # print(f"Total sales: ${highest_sales:.2f}")
16 # import pandas as pd
17
18 ## Prompt the user for the file name
19 file_name = input()
20
21 ## Load the data
22 df = pd.read_csv(file_name)
23
24
25 ## Find the month with the highest total sales
```

5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.05,,S
```

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Scatter Plot for Age vs. Fare by Survived
17 colors = data['Survived'].map({0: 'red', 1: 'blue'})
18 plt.scatter(data['Age'], data['Fare'], c=colors)
19 plt.title('Age vs. Fare by Survival')
20 plt.xlabel('Age')
21 plt.ylabel('Fare')
22 plt.show()
23
24
25
```

5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
```

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Fare by Pclass
17 plt.figure()
18 plt.scatter(data['Age'], data['Fare'])
19 plt.title('Age vs. Fare')
20 plt.xlabel('Age')
21 plt.ylabel('Fare')
22 plt.show()
23
24
25
```

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5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

- Use the **Pclass** column to group the data for the boxplot.
- Set the title of the plot to **"Fare by Pclass"**.
- Remove the default subtitle with **plt.suptitle("")**.
- Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S,
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S,
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, S,
6, 0, 3, "Moran, Mr. James", male, 0, 0, 0, 330877, 8.4583, Q,
7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
```

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Fare by Pclass
17 plt.figure(figsize=(8,6))
18 data.boxplot(column='Fare', by='Pclass')
19 plt.title('Fare by Pclass')
20 plt.suptitle('')
21 plt.xlabel('Pclass')
22 plt.ylabel('Fare')
23 plt.show()
24
25
```

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5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

- Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
- Set the title of the plot to **"Age by Survival"**.
- Remove the default subtitle with **plt.suptitle("")**.
- Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S,
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S,
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, S,
6, 0, 3, "Moran, Mr. James", male, 0, 0, 0, 330877, 8.4583, Q,
```

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Survived
17 plt.figure(figsize=(8,6))
18 data.boxplot(column='Age', by='Survived')
19 plt.title('Age by Survival')
20 plt.suptitle('')
21 plt.xlabel('Survived')
22 plt.ylabel('Age')
23 plt.show()
24
25
```

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5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
```

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Survived
17 plt.figure(figsize=(8,6))
18 data.boxplot(column='Age', by='Survived')
19 plt.title('Age by Survival')
20 plt.suptitle('')
21 plt.xlabel('Survived')
22 plt.ylabel('Age')
23 plt.show()
24
25
```

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5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Wilson, Mr. Charles	male	30	0	0	349000	51.87	C	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Wilson, Mr. Charles",male,30,0,0,349000,51.87,C
```

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Pclass
17
18 data.boxplot(column = 'Age', by='Pclass')
19 plt.title('Age by Pclass')
20 plt.suptitle('')
21 plt.xlabel('Pclass')
22 plt.ylabel('Age')
23 plt.show()
24
25
```

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5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title **"Survival by Embarked"** to the chart.
4. Label the x-axis as **'Embarked'** and the y-axis as **'Count'**.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as **'Survived'** and **'Not Survived'**).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Embarked
17 survival_counts = data.groupby('Embarked_Q')
18 ['Survived'].value_counts().unstack().fillna(0)
19 survival_counts.plot(kind='bar', stacked=True)
20 plt.title('Survival by Embarked')
21 plt.xlabel('Embarked')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
```

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5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Pclass"** to the chart.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Pclass
17 survival_counts = data.groupby('Pclass')
18 ['Survived'].value_counts().unstack().fillna(0)
19 survival_counts.plot(kind='bar', stacked=True)
20 plt.title('Survival by Pclass')
21 plt.xlabel('Pclass')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
```

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5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

- Use the **'Survived'** column to show the count of survivors (0 = Did not survive, 1 = Survived).
- Set the chart type to **'bar'**.
- Add the title **'Survival Count'** to the chart.
- Label the x-axis as **'Survived'** and the y-axis as **'Count'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171,7.25,,S			
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71.2833,C85,C			
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7.925,,S			
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53.1,C123,S			
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8.05,,S			
6	0	3	Moran, Mr. James	male	0	0	0	330877,8.4583,,Q			
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463,51.8625,E46,S			
8	0	3	Wilson, Mr. John	male	29	0	0	349000,51.076,,C			

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Wilson, Mr. John",male,29,0,0,349000,51.076,,C
```

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival Rate
17 survival_counts = data['Survived'].value_counts()
18 survival_counts.plot(kind='bar')
19 plt.title('Survival Count')
20 plt.xlabel('Survived')
21 plt.ylabel('Count')
22 plt.show()
23
24
25
```

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5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use **30 bins** for the histogram.
- Set the **edge color** of the bars to **black** (k).
- Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
- Add the title **'Age Distribution'** to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171,7.25,,S			
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71.2833,C85,C			
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7.925,,S			
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53.1,C123,S			
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8.05,,S			
6	0	3	Moran, Mr. James	male	0	0	0	330877,8.4583,,Q			
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463,51.8625,E46,S			
8	0	3	Wilson, Mr. John	male	29	0	0	349000,51.076,,C			

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Wilson, Mr. John",male,29,0,0,349000,51.076,,C
```

Histogram...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17 plt.hist(data['Age'],bins=30,edgecolor='k')
18 plt.title('Age Distribution')
19 plt.xlabel('Age')
20 plt.ylabel('Frequency')
21 plt.show()
22
23
```

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